



MASTER IN ASTROPHYSICS  
AND SPACE SCIENCE



УНИВЕРЗИТЕТ У БЕОГРАДУ  
UNIVERSITY OF BELGRADE

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# Nuclear Starburst Disks

ACTIVE GALACTIC NUCLEI

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- $10^{5.5-6.5} M_{\odot}$  Mass Systems



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- Extended Emission Line Regions (Short Lived SMBH Flare Ionization)



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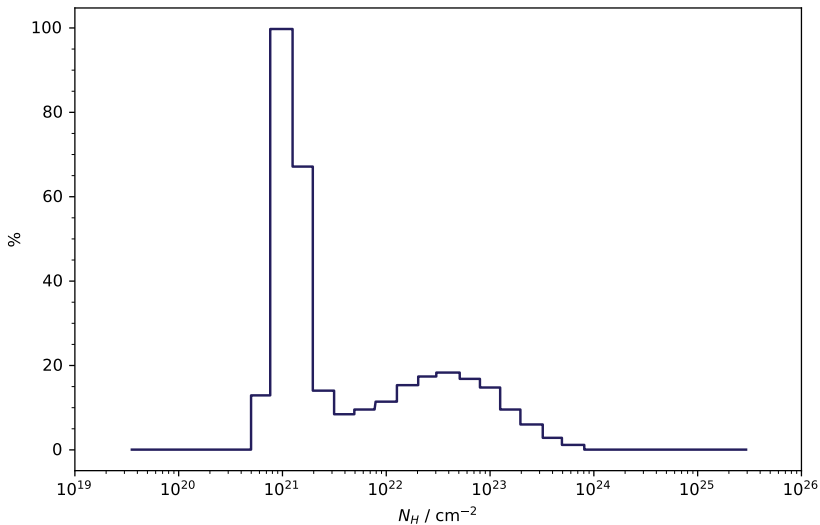
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- Extreme Mass Ratio Inspiral (SMO/BH EMRI)



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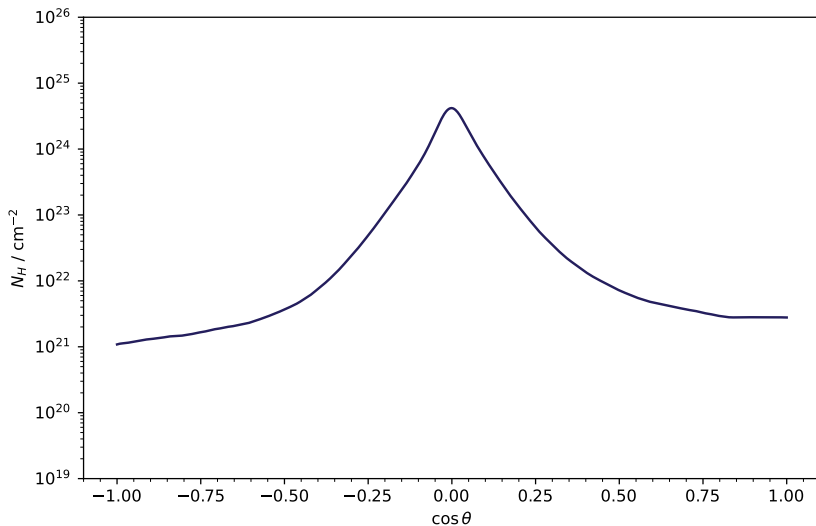


Modelling

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## Angular Obscuring

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## Outlook

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- Precision Cosmology Measurements

# Thank You!

- E. Kara & J. García, *Supermassive Black Holes in X-Rays: Standard Accretion to Extreme Transients* (2025) arXiv:2503.22791 [astro-ph.HE]
- A. Franchini et. al., *Quasi Periodic Eruptions from Impacts with Rigidly Precessing Accretion Disc in Extreme Mass Ratio Inspiral Systems* (2023) A&A Volume 675 Article 100

Resources on GitHub: [here](#)